Promo Finder: finding 20-60s promos in microdrama screenplays

Kuku AI Generalist assignment · Problem 1 (Find the Promos) · code, raw outputs and app in the repo · results below are from the 50-episode sample show *I Married a Broken Soldier* (102.7 min)

1. The problem and why this one

Today an editor watches every episode of every new show to pick promo moments, and a new show ships every week, so the work grows with the catalogue and blocks paid-social launch. It is a good fit for AI because the screenplay already holds what matters: every line and action with a start and end time. A model can read 50 episodes in minutes and judge drama, while code can make every timestamp exact. The output is also easy to check: a clip either opens on a hook, ends unresolved and makes sense to a stranger, or it does not.

2. Assumptions

- Screenplay times match the edited episode (or a constant offset), so script time ranges are the cut points. Clips are unedited footage: segments only in story order, at most 3 moments, total 20-60s.
- "Hook in the first 3 seconds" = the hook line or action *starts* within 3s of the clip start. Visual-only beats are judged from the CAMERA/ACTION text, since there is no footage at this stage.
- Someone who taps a promo usually starts at episode 1, so early-episode promos get a small bonus and late ones are treated as spoiler risk. The tool finds and ranks promos; a human approves before money is spent.

3. How it works: AI vs code vs human

Step	Who	What it does
Parse	code	Scenes and 3,700 timed events; every line gets an ID like **5:024** (episode:line).
Show bible	AI ×1	Maps inconsistent speaker labels ("LADY IN GOLD GOWN" = Clarissa); premise; list of twists with the episode each is revealed in (used to flag spoilers).
Propose	AI	Batches of 3 episodes (+ the next as context): 4 candidates per episode, named as **line IDs, never timestamps**, plus hook, cliffhanger, why, trope.
Validate + repair	code → AI ×1	IDs → exact times, cuts snapped to shot boundaries without chopping a spoken line; checks 20-60s, hook within 3s, cliffhanger is the last beat, the next line does not leak in. Failures go back to the model once with the exact error.
Cold viewer	AI (cheap)	A separate call sees **only the clip**, no show context, and reports what it understood, its open question, whether the ending felt finished, and whether it would tap.
Judge	AI	Anchored 1-10 rubric (hook, cliffhanger, standalone, emotion, spoiler) with the script and the cold-viewer report; also says whether the clip already answers its own question.
Score + rank	code	Both scorers turned into percentiles within the show, then weighted (30/30/25/15). The cliffhanger is capped if either says the ending resolves; a penalty applies if both say a stranger would not follow it. Then spoiler/rule penalties, an episode-1 bonus, de-duplication and variety. Confidence = how closely the two scorers agree.
Launch plan	AI ×1	Picks the first 10 to post and their order (strength, trope variety, funnel fit).
Approve	human	Web app (Flask): the 10 to post first, each with where to cut, approve/reject, the script on demand; CSV export.

Why this split: models are good at judging drama and bad at arithmetic on timestamps, so the model never writes a number. Standalone is *tested* by a viewer with no context rather than asserted by the model that read the whole show. Every model call goes through one module; swapping models is one line in **config.toml** or a CLI flag, and any OpenAI-compatible API works.

4. Output for the sample show: first five to post

#	Promo	Segments (start-end)	Hook (first 3s)	Cliffhanger (cut)	Why it works
1	E31-4 Burning the Grimoire	Ep31 02:10.8-02:35.2 (24s)	'Enough!' - a red-hot iron thrust into her mother's book	The iron swings up and stops millimetres from her eye	Fast escalation; a visual cliffhanger needing no context
2	E15-3 Blood and Chamomile	Ep15 00:49.5-01:26.0 (37s)	Blood through her bandages as thugs crush her herbs	"You want to fight us, veteran?" as he steps in	Victim-to-protector setup; cut before the fight
3	E05-1 Stripped of Everything	Ep5 00:15.8-00:58.5 (43s)	"Your dowry is revoked. Your inheritance is revoked."	Her dead mother's ledger flung into the mud	Pure injustice in episode 5: early funnel fit
4	E03-4 The Proposal	Ep3 01:59.2-02:25.8 (27s)	"You came here to become a wife. They took that from you."	"Lady Juliana... accept my hand." (before her answer)	Wish-fulfilment beat; tests romance appetite
5	E32-4 Chained and Fearless	Ep32 02:05.4-02:34.4 (29s)	Chained, she mocks her captor's cowardice	Poker glowing at her unblinking eye	Defiance + peril (but see failures: same set piece as #1)

All 40 ranked promos (each with segments, hook, cliffhanger, why, scores and confidence), every candidate, and every raw prompt and reply are in **outputs/final/**.

5. Scale and cost per show

	Per 50-episode show (103 min of content)
Model calls	276: 1 bible, 22 propose, 11 repair, 218 cold-viewer, 23 judge (incl. retries for malformed replies), 1 plan. About 1.1M input + 0.4M output tokens (output includes the model's thinking).
Cost (list price)	≈ **$4.80** (about $0.10 per episode) on Gemini 3.5 Flash with 3.1 Flash-Lite for the cold viewer; ≈ $1.25 with Flash-Lite everywhere. Two-thirds of the cost is 3.5 Flash output tokens in propose/repair. Batch API pricing would halve it.
Time	24 min on a free key (throttled to 9-14 requests/min); a rerun from cache takes seconds. On a paid tier with more workers, a few minutes. Editor time drops from watching 103 min to reviewing ~10-20 clips in the app.
Limits seen	The free tier allows ~20 requests/day on 3.5 Flash, so a free-key run finishes on fallback models (3 Flash preview, 3.5 Flash-Lite); the run log records which. For weekly production, use a paid key and **episodes_per_call = 1** (see failures).

6. How we know it is good

A reference set of 18 promos across 12 episodes was drafted by reading the scripts, using Claude (a different model family, so the Gemini pipeline is not grading itself), then checked by hand. A match = time overlap IoU ≥ 0.35, or the core line plus IoU ≥ 0.20. The

yardstick is a no-AI keyword picker run through the same validator.

Metric (sample show)	Final	First run today	Keyword baseline
Reference promos found among candidates	18/18	18/18	14/18
... cut like an editor would (start and end within 3s)	7/18	7/18	2/18
... that made the ranked top 40	11/18	13/18	-
Top 10 whose ending already resolves (per judge or cold viewer)	**0/10**	4/10	-
Ranked promos in reference episodes that match a reference	12/16	13/17	-
Confidence mix of ranked 40 (high / medium / low)	0 / 21 / 19	0 / 5 / 35	-
Proposals passing every code rule first time; fixed by repair	84%; 54	84%	-
Top 10 that reappear in another run's top 20	6-7/10	-	-

Reading the first run's top picks by hand changed the scoring. #1 was a speech that finishes on its point, and #4 cut *after* the villain had fled. The judge gave them cliffhanger 8 and 9, while the cold viewer rated need-to-know 3 for both. Fixes: both scorers now say explicitly whether the clip answers its own question (if so, the cliffhanger is capped), scores are compared as percentiles because the two models use the scale differently, and a promo neither scorer can follow is penalised. Ranked recall fell from 13 to 11 of 18: some reference promos end on a reveal that the scorers call resolved. I tested a softer cap; it lost more (9/18), so I kept the hard cap.

Where it still picks badly (honest failures)

- **Batch-relative judging.** The same clip scored 8/7/7/8 when judged alongside episodes 4-6 and 5/5/5/6 alongside 7-9, so a good promo (the husband's silent arrival, ep 7) fell out of the top 40. Fix: one episode per judge call, or a stronger judge.
- **The cheap cold viewer is trigger-happy.** It calls 54% of clips "finished" against 35% for the judge, and they agree on only 57%. Judge-vs-viewer correlation is r = 0.41 for standalone and 0.27 for hook, so confidence is never 'high'.
- **Mid-show set pieces dominate.** Launch picks #1 and #5 (ranked #1 and #2) are consecutive beats of the same hot-poker scene (eps 31-32), and an episode-50 clip ranks #4 despite the funnel bonus. Duplicate detection only compares time overlap, not story beats.
- **Episode 1 is under-served.** The altar betrayal ("Because he is marrying me") ranks only #26: the cold viewer feels the reveal answers itself. An editor would post it first.
- Visual hooks are judged from text. A clip whose first 3s are camera-only is flagged, but nobody has watched the footage.

7. AI tools used, and one place AI got it wrong

In the product: Gemini 3.5 Flash (bible, propose, judge, plan), Gemini 3.1 Flash-Lite (cold viewer), with 3 Flash / 3.5 Flash-Lite as automatic fallbacks. **To build it:** Claude wrote and tested the code with me, drafted the reference set (checked by hand), and in the final session ran the show, read the outputs, and made the fixes above.

Where AI got it wrong. The judge model gave the first run's #1 promo a cliffhanger score of 8 while its own note said the clip *"resolves her speech fully instead of cutting one line earlier"*. The model could see the flaw but did not let it change the number. That is why the final design asks each scorer a yes/no question about the ending, and lets code, not the model, apply the consequence. The AI-written code also slipped: it treated any rate-limit reply that mentioned a per-day quota as "daily quota exhausted" and silently moved to weaker models. It also dropped valid repaired promos the first time I tightened the repair step (this cost reference promo G27b until caught by the eval). Both are fixed, and fallbacks are now logged.

8. What I would do next

- Judge one episode per call on a paid key (removes the batch effect above; about +$1 per show).
- Calibrate the weights on real data: post the launch plan, then fit the score to actual 3-second hold rate and tap-through. The anchored rubric is a guess until it meets real CTR.
- Beat-level de-duplication (same characters + same stakes across adjacent episodes), and a guaranteed slot for the best episode-1 promo.
- Once footage exists, check the first 3 seconds with a video model, and map script times to the edit if they drift.